USE CASE SPECIFICATION

Online Internship & Placement Prep Portal

UC-1: Register & Login

Actor(s)	Student, Admin
Goal	Allows a new user to create an account and an existing user to authenticate and reach the dashboard for their role.
Pre-condition	The user has access to the portal's landing page. For login, the user already has a registered account.
Main Flow	**1.** User navigates to the portal's landing page. **2.** User selects **Register** (new user) or **Login** (existing user). **3.** For registration, user enters name, email, password, and role-specific details (e.g., college/batch for students). **4.** System validates input format (email syntax, password strength). **5.** System checks that the email is not already registered. **6.** System creates the account and stores the password in hashed form. **7.** System redirects the user to the login page with a success message. **8.** User enters email/username and password on the login page. **9.** System validates credentials against the stored hashed values. **10.** System creates an authenticated session and redirects the user to their respective dashboard (Student or Admin).
Alternate Flow	**Invalid Login Credentials** At Step 9, if the credentials do not match: **9a.** System displays an **Invalid email or password** message. **9b.** System returns the user to the login form (**Step 8**) without creating a session. **Duplicate Registration Email** At Step 5, if the email is already registered: **5a.** System displays an **Account already exists** message. **5b.** System offers a link to the login page instead of creating a new account.
Post-condition	User is registered (if new), authenticated, and directed to the appropriate Student or Admin dashboard.

UC-2: View Internship Listings

Actor(s)	Student
Goal	Allows a student to browse, filter, and view details of active internship listings posted by the admin.
Pre-condition	The student is logged in (authenticated).
Main Flow	**1.** Authenticated student navigates to the **Internships** section. **2.** System retrieves active internship listings posted by admin from the database. **3.** System displays listings with title, company, and short description. **4.** Student applies filters (location, domain, stipend, duration) as needed. **5.** System re-queries and refreshes the listing view based on the filters. **6.** Student selects a specific listing to view its details. **7.** System displays the full description, eligibility, deadline, and current application status (if already applied). **8.** Student may bookmark/save a listing for later reference. **9.** Student navigates back to the full listings page. **10.** Student proceeds to apply directly from the detail view, triggering the **Apply for Internship** use case.
Alternate Flow	**No Listings Available** At Step 3, if no active listings exist: **3a.** System displays **No internships currently available.** **Filter Returns No Results** At Step 5, if the filter criteria match no listings: **5a.** System displays **No matching internships found.** **5b.** System lets the student reset the filters.
Post-condition	Student has viewed the listing details and, optionally, saved a listing or proceeded to apply.

UC-3: Apply for Internship

Actor(s)	Student
Goal	Allows a student to submit an application, with resume and cover note, for an active internship listing. (Includes: View Internship Listings)
Pre-condition	The student is logged in and the internship listing is active.
Main Flow	**1.** Student views internship listings (**include:** View Internship Listings). **2.** Student selects a listing and clicks **Apply**. **3.** System checks whether the student has already applied to this listing. **4.** System displays the application form (resume upload, cover note, etc.) if not already applied. **5.** Student fills in / reviews the required application details. **6.** Student submits the application. **7.** System validates that all mandatory fields and attachments are present. **8.** System records the application with a timestamp and an **Applied** status. **9.** System updates the listing's application count for the admin view. **10.** System displays a confirmation and adds the entry under **My Applications**.
Alternate Flow	**Duplicate Application Blocked** At Step 3, if the student has already applied: **3a.** System blocks the action. **3b.** System displays **You have already applied to this internship** instead of showing the form. **Incomplete Application** At Step 7, if mandatory fields are missing: **7a.** System highlights the missing fields. **7b.** System returns to **Step 5** without submitting.
Post-condition	Application is recorded with "Applied" status, listed under My Applications, and visible to the admin.

UC-4: Practice Questions (Aptitude / MCQ / Coding)

Actor(s)	Student
Goal	Allows a student to practise questions by category and receive immediate feedback, with the session results stored for analytics.
Pre-condition	The student is logged in and the question bank contains questions for the chosen category.
Main Flow	**1.** Student navigates to the **Practice** section. **2.** System displays available categories (Aptitude, Technical MCQ, Coding). **3.** Student selects a category. **4.** System retrieves questions from the categorized question bank matching that category. **5.** System displays the first question with answer options / code editor as applicable. **6.** Student submits an answer for the current question. **7.** System evaluates the answer and shows immediate correctness feedback. **8.** Student proceeds to the next question (**Steps 6–7** repeat until the set is exhausted or the student stops). **9.** System compiles the practice session score once the student completes or ends the set. **10.** System stores the session results and updates the student's practice history for analytics.
Alternate Flow	**No Questions in Category** At Step 4, if the question bank has no questions for the selected category: **4a.** System displays **No questions available in this category yet.** **4b.** System returns to **Step 2**. **Student Exits Mid-Session** At Step 8, if the student exits before completing the set: **8a.** System saves partial progress. **8b.** System stores results up to the last answered question instead of discarding the session.
Post-condition	Practice session score and results are stored in the student's practice history.

UC-5: Take Timed Mock Test

Actor(s)	Student
Goal	Allows a student to attempt a timed mock test and have it scored and stored in their history. (Includes: Auto-Submit on Timeout, View Score & Answer Analysis)
Pre-condition	The student is logged in and at least one mock test is available.
Main Flow	**1.** Student navigates to the **Mock Tests** section. **2.** Student selects a mock test (by category, difficulty, or duration). **3.** System displays the test instructions and time limit. **4.** Student clicks **Start Test** and the system starts the countdown timer. **5.** System presents questions with navigation controls to move between them. **6.** Student answers questions, marking/reviewing/changing answers within the time limit. **7.** Student clicks **Submit Test** before time expires. **8.** System locks further input and records all submitted answers. **9.** System calculates the score and correct/incorrect breakdown (**include:** View Score & Answer Analysis). **10.** System stores the test attempt in the student's history for analytics.
Alternate Flow	**Time Expires Before Submission** At Step 7, if the timer reaches zero before manual submission: **7a.** System automatically submits the test using the recorded answers (**include:** Auto-Submit on Timeout). **7b.** System proceeds to **Step 8**. **Connectivity Loss During Test** At Step 6, if the student's connection drops: **6a.** System preserves the already-recorded answers. **6b.** System resumes the same session with the remaining time when the student reconnects, instead of restarting the test.
Post-condition	Test attempt is submitted, scored, and stored in the student's test history.

UC-6: Auto-Submit on Timeout

Actor(s)	System (triggered on behalf of Student)
Goal	Automatically submits a mock test when the time limit expires so that the attempt is saved and scored fairly.
Pre-condition	A mock test attempt is in progress and the countdown timer is running.
Main Flow	**1.** System continuously tracks elapsed time against the test's configured duration. **2.** Timer reaches zero while the student is still attempting the test. **3.** System immediately disables all answer input controls. **4.** System captures the current state of all answered and unanswered questions. **5.** System marks the test attempt status as **Auto-Submitted**. **6.** System records the submission timestamp as the configured end time. **7.** System processes the recorded answers the same way as a manual submission. **8.** System notifies the student on-screen that the test was auto-submitted due to timeout. **9.** System proceeds to scoring and analysis generation. **10.** System logs the auto-submit event for audit and analytics purposes.
Alternate Flow	**Student Had Just Manually Submitted** At Step 2, if the attempt already has a "Submitted" status when the timer expires (race condition): **2a.** System skips the auto-submit process. **2b.** System avoids creating a duplicate record.
Post-condition	Test attempt is saved with "Auto-Submitted" status and sent for scoring, and the event is logged.

UC-7: View Score & Answer Analysis

Actor(s)	Student
Goal	Allows a student to see the score, accuracy, and a question-by-question analysis of a completed test attempt.
Pre-condition	The student is logged in and has submitted at least one mock test attempt.
Main Flow	**1.** Student completes/submits a mock test, or navigates from **My Test History**. **2.** System retrieves the recorded answers for the selected attempt. **3.** System compares each recorded answer against the correct answer key. **4.** System computes the total score, accuracy percentage, and time taken. **5.** System displays a results summary (score, rank/percentile if applicable). **6.** Student clicks into **Detailed Analysis**. **7.** System displays a question-by-question breakdown (correct / incorrect / skipped) with explanations. **8.** Student filters the breakdown by topic or difficulty. **9.** Student reviews individual questions and their correct solutions. **10.** Student navigates back to test history or proceeds to the analytics dashboard.
Alternate Flow	**Analysis Not Yet Generated** At Step 3, if scoring is still being computed (e.g., delayed evaluation for coding questions): **3a.** System displays **Results are being processed.** **3b.** Student can check back later.
Post-condition	Student has reviewed the score summary and detailed answer analysis for the attempt.

UC-8: View Personal Analytics Dashboard

Actor(s)	Student
Goal	Allows a student to track performance trends and identify strengths and weaknesses across tests and practice sessions. (Includes: View Score & Answer Analysis)
Pre-condition	The student is logged in.
Main Flow	**1.** Student navigates to the **Analytics / Dashboard** section. **2.** System retrieves the student's full test and practice history. **3.** System aggregates scores across attempts to compute performance trends over time. **4.** System identifies strength and weakness areas by topic/category. **5.** System renders trend charts (e.g., score over time, accuracy by category). **6.** Student drills down into a specific test attempt for full details (**include:** View Score & Answer Analysis). **7.** Student filters the dashboard by date range or test type. **8.** System recalculates and refreshes the visualizations based on the filter. **9.** Student views recommended focus areas generated from the weakness analysis. **10.** Dashboard auto-updates the next time the student completes a new test or practice session.
Alternate Flow	**Insufficient History** At Step 3, if the student has no completed tests or practice sessions yet: **3a.** System displays **Complete a test or practice session to see your analytics** instead of charts.
Post-condition	Student has viewed up-to-date performance trends and recommended focus areas.

UC-9: Manage Internship Listings

Actor(s)	Admin
Goal	Allows an admin to add, edit, and deactivate internship listings that are shown to students.
Pre-condition	The admin is logged in.
Main Flow	**1.** Admin navigates to **Manage Internships**. **2.** System displays existing listings with status (active / expired / draft). **3.** Admin selects **Add New Listing**. **4.** Admin enters listing details (title, company, description, eligibility, deadline, stipend). **5.** Admin submits the listing. **6.** System validates required fields and deadline validity (must be a future date). **7.** System saves the listing and marks it **Active**, making it visible to students. **8.** Admin selects an existing listing to edit its details. **9.** Admin saves changes; system updates the listing and timestamps the modification. **10.** Admin deactivates/removes a listing; system hides it from student view while retaining historical application records.
Alternate Flow	**Validation Failure** At Step 6, if required fields are missing or the deadline is in the past: **6a.** System rejects the submission and highlights the errors. **6b.** System returns to **Step 4**. **Removing a Listing with Active Applications** At Step 10, if the listing has pending student applications: **10a.** System warns the admin. **10b.** System asks for confirmation before deactivating the listing.
Post-condition	Listings are added, updated, or deactivated and students see only the active ones.

UC-10: Manage Categorized Question Bank

Actor(s)	Admin
Goal	Allows an admin to add, edit, and delete categorized questions used for practice and mock tests.
Pre-condition	The admin is logged in.
Main Flow	**1.** Admin navigates to the **Question Bank** section. **2.** System displays existing categories (Aptitude, Technical MCQ, Coding) with question counts. **3.** Admin selects a category or creates a new one. **4.** Admin selects **Add Question**. **5.** Admin enters question content, options/test cases, correct answer, and difficulty level. **6.** Admin submits the question. **7.** System validates that all required fields (e.g., correct answer, options for MCQ) are present. **8.** System saves the question to the selected category, making it available for practice and tests. **9.** Admin edits or deletes existing questions from the bank. **10.** System updates the question bank and reflects the change immediately.
Alternate Flow	**Incomplete Question Data** At Step 7, if a required field (e.g., no correct answer marked) is missing: **7a.** System rejects the submission. **7b.** System returns to **Step 5** with the error highlighted. **Deleting a Question in Use** At Step 9, if the question is part of an active or scheduled mock test: **9a.** System warns the admin before allowing the deletion.
Post-condition	Question bank is updated and the changes are available to practice sessions and mock tests.

UC-11: View Student Applications

Actor(s)	Admin
Goal	Allows an admin to review student applications across all listings and update their status.
Pre-condition	The admin is logged in.
Main Flow	**1.** Admin navigates to the **Applications** section. **2.** System retrieves all applications submitted across all internship listings. **3.** System displays applications in a table with student name, listing, and status. **4.** Admin filters applications by listing, date, or status. **5.** System refreshes the table based on the filter. **6.** Admin selects a specific application to view full details. **7.** System displays the student's submitted resume/cover note and profile information. **8.** Admin updates the application status (e.g., Shortlisted, Rejected, Selected). **9.** System saves the status change and timestamps it. **10.** System reflects the updated status to the student under **My Applications**.
Alternate Flow	**No Applications Yet** At Step 3, if no student has applied to any listing: **3a.** System displays **No applications received yet.** **Bulk Status Update** At Step 8, if the admin wants to update several applications at once: **8a.** Admin selects multiple applications and applies a status change in bulk. **8b.** System validates and updates all selected records in a single operation.
Post-condition	Application status is updated, timestamped, and visible to the student.